Shubhi Rastogi Embedded Developer

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Summary

Embedded Systems Engineer with 5+ years of experience in automotive ECU firmware development, safety-critical RTOS environments and ARM Cortex architecture. Strong expertise in Embedded C, CAN/UDS diagnostics, secure bootloader design, motor control firmware, board bring-up, OTA firmware updates, telematics, and production issue debugging. Hands-on experience with embedded security concepts including secure boot, firmware integrity verification, and rollback-safe update mechanisms. Proven track record of improving firmware stability, reducing field failures, and optimizing system reliability in high-volume vehicle platforms.

Skills

Microcontrollers:	NXP S32K3xx, Renesas RL78, TI C2000, Quectel EC200, STM , ARM cortex architecture
Protocols:	CAN, UDS, MQTT, UART, SPI, I2C
Driver Development:	Low-level peripheral drivers (CAN, UART, SPI, I2C, ADC, PWM, Timers)
Languages:	Embedded C, C, Python
Tools:	Git, CANoe, CANalyzer, Wireshark, Jira, logic analyzers
Debugging:	JTAG, UART debugging, CAN trace analysis
Peripherals:	PWM, ADC, Timers
OS:	FreeRTOS

Experience

Embedded Developer — Ola Electric, Bengaluru *Dec 2023 — Present*

- Enhanced BCM firmware by integrating UART-based Bluetooth interface for customer app connectivity, improving communication stability and reducing app connection failures by 15%.
- Led board bring-up activities including IMU integration, watchdog configuration, and reset-reason diagnostics, contributing to 20% reduction in unexpected system crashes.
- Designed and optimized CAN-to-UART gateway on MSPM0 architecture using circular buffer implementation, improving data handling reliability and reducing data-loss scenarios under high-load conditions.
- Implemented secure bootloader for instrument cluster enabling firmware update and recovery mechanisms, improving firmware update success rate and reducing field reflash dependency by 30%.
- Collaborated with hardware, cloud, and validation teams to ensure end-to-end ECU integration across vehicle platforms.
- Developed IoT-connected telematics features including parental control logic, backend API integration, and COTA (Configuration Over-the-Air) setup — enabling remote device configuration and reducing manual service intervention.
- Developed STM32 based CAN Diagnostic tool for diagnosis.
- Resolved production and field issues using cloud diagnostics, CAN trace analysis, and hardware-level debugging, improving issue resolution turnaround time by 25%.

Embedded Firmware Engineer — Simple Energy, Bengaluru *Aug 2023 — Dec 2023*

- Maintained and enhanced BCM firmware for vehicle production releases, improving integration stability and reducing validation rework across production builds.
- Supported end-to-end subsystem integration and validation across multiple ECUs, reducing integration issues during production releases.

- Contributed to CAN communication stack validation and DBC workflow refinement, improving message validation accuracy.

Software Designer — Alstom Transport Ltd. *Mar 2021 — Aug 2023*

- Developed and modified embedded software modules in a safety-critical RTOS environment based on evolving system requirements.
- Improved code maintainability by reducing cyclomatic complexity by 18% and implementing defensive programming techniques, lowering defect leakage in production builds.
- Automated log analysis workflows using Python, reducing manual debugging effort by approximately 25% and improving issue resolution turnaround time.
- Worked within railway signaling systems following structured change management and quality processes aligned with safety-critical standards.

Education

PG Diploma — Embedded System Design (CDAC ACTS, Bangalore) *2020 — 2021*
B.Tech — Electronics & Communication *2015 — 2019*

Certifications

- HackerRank Python (Basic)
- HackerRank Problem Solving